A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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Abstract

Large language model (LLM) calibration is nearly always reported as a single aggregate scalar, obscuring whether miscalibration is diffuse or concentrated in specific knowledge domains. We develop a statistically rigorous audit framework and apply it to MMLU [Hendrycks et al., 2021] using token-level log probabilities from seven instruction-tuned models spanning three families: GPT-4o-mini, Llama-3-8B-Instruct, and Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. Between-subject calibration variance accounts for 1.7–7.9% of the total across models. Subject-level ECE rankings are highly correlated for the six larger models (Spearman $\rho \in [0.80, 0.94]$, all 15 pairs); Qwen-2.5-0.5B is the exception, showing extreme positional bias that produces a qualitatively different pattern. The same subjects are robustly worst-calibrated (college-level quantitative STEM, niche factual recall, structured legal/professional reasoning) and robustly best-calibrated (high-school-level Social Sciences and applied/factual subjects). Within the Qwen family, the entropy coefficient grows monotonically with scale ($-0.004 \rightarrow +0.006 \rightarrow +0.029 \rightarrow +0.040$, last two $p < 0.001$), consistent with prior reports of RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025].

1 Introduction

Large language models are increasingly deployed in high-stakes settings—healthcare, law, education—where the reliability of model-expressed confidence is as important as accuracy itself. A well-calibrated model is one whose stated confidence tracks its empirical accuracy: if a model assigns 80% probability to an answer, it should be correct 80% of the time. Miscalibration, particularly overconfidence, means that users who rely on model confidence as a decision signal will be systematically misled.

A substantial literature has studied LLM calibration at the aggregate level; we survey it in Section 2. However, to our knowledge, no existing work provides a formal *statistical audit* of that heterogeneity. Specifically, the existing literature reports subject-level ECE without (i) stabilizing estimates against large differences in per-subject sample size, (ii) formally testing which subjects are robustly miscalibrated after correcting for multiple comparisons, (iii) decomposing how much total calibration variance is attributable to domain versus subject versus individual question, or (iv) characterizing subjects by their *pattern* of miscalibration rather than its scalar magnitude.

This framing is analogous to the gap between per-group accuracy reporting and a rigorous fairness audit [Barocas et al., 2023]: the fairness literature long ago recognized that single-number summaries are insufficient for subgroup auditing, and we import that logic into calibration evaluation. The downstream implication of the audit is structural: if miscalibration is diffuse, a single global recalibration (e.g., temperature scaling) is a reasonable fix; if it is concentrated in specific subject clusters and those clusters are stable across models, then subject-aware recalibration becomes a structurally appropriate scoping choice. Our contribution is the diagnostic arm: stable audit findings establish whether the structural assumption behind a subject-aware correction is warranted in the first place.

Our central scientific questions, restated:

1. **Where** is calibration good and bad? Variance decomposition, FDR-corrected subject rankings, and a within-MMLU contrast between well-calibrated and poorly-calibrated subject clusters.

2. **How consistent** is the answer across models? Pairwise Spearman correlations of subject-level ECE across seven models.

3. **What mechanisms** produce the observed heterogeneity? Question- level predictors, plus a within-Qwen scaling sweep across four matched- precision checkpoints (1.5B, 3B, 7B, 14B).

2 Related Work

Calibration of neural networks. Guo et al. [2017] showed that modern neural networks are systematically overconfident and that temperature scaling largely corrects this at the aggregate level, with Zadrozny and Elkan [2002] earlier establishing isotonic regression as an alternative nonparametric recalibration method. Kadavath et al. [2022] extended calibration evaluation to LLMs, establishing that token-level probabilities on multiple-choice tasks carry a meaningful calibration signal. Nakkiran et al. [2025] and Yaldiz et al. [2026] study how instruction tuning affects calibration, finding that RLHF-style fine-tuning tends to degrade calibration relative to base models, an effect also documented in Ouyang et al. [2022].

Domain-level heterogeneity. Xiong et al. [2024] is the closest prior work in spirit, benchmarking confidence elicitation across task types. Jiang et al. [2021] showed that question-level features predict calibration on QA tasks. Tan et al. [2026] uses base model signals for unsupervised recalibration. None of these works performs a formal statistical audit of subject-level heterogeneity.

Multiple testing and multilevel modeling. The Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] controls the false discovery rate and is standard in high-dimensional inference. We use the multilevel modeling framework of Raudenbush and Bryk [2002] to decompose calibration variance across levels of the MMLU hierarchy.

3 Data

MMLU. We use the Massive Multitask Language Understanding benchmark [Hendrycks et al., 2021], accessed via the `cais/mmlu` dataset on HuggingFace. MMLU comprises 14,042 four-option multiple-choice questions across 57 subjects nested in four broad domains: STEM, Social Sciences, Humanities, and Other. Subject sizes range from 100 to 1,534 questions (median 173). The three-level hierarchy—questions $\rightarrow$ subjects $\rightarrow$ domains—is natural for the proposed analysis. We use the test split throughout.

Models. We query seven instruction-tuned models spanning three families and six parameter scales. **GPT-4o-mini** is queried via the OpenAI Batch API at temperature 0 (`max_tokens=1`, `top_logprobs=20`); the choice of 20 (rather than the API default of 5) ensures all four answer tokens are captured for virtually every question. **Llama-3-8B-Instruct** and **Qwen-2.5 {0.5B,**

1.5B, 3B, 7B, 14B}-Instruct are run locally on a Colab GPU. The 3B, 7B, 14B, and Llama (8B) models use 4-bit quantization (BitsAndBytes; Dettmers et al. 2023); the 0.5B model also uses 4-bit, and the 1.5B model is run twice (once at 4-bit, once at fp16) so the precision effect can be examined directly.[1] For all seven models we pre-fill the assistant turn with "The answer is" before extracting logprobs at the next-token position. Test-set accuracies are 0.749 (GPT-4o-mini), 0.606 (Llama), 0.405 (Qwen-0.5B), 0.571 (Qwen-1.5B fp16), 0.601 (Qwen-3B), 0.685 (Qwen-7B), 0.757 (Qwen-14B). Qwen-0.5B exhibits extreme positional bias (predicting D 59% of the time vs. A 5.6%), discussed where it affects results.

4 Methods

4.1 Confidence Elicitation and Outcome

For each question, the model's confidence $c_i \in (0, 1]$ is the maximum of the softmax-renormalized distribution over the four answer tokens A/B/C/D, equal to $P(\hat{y} \mid \hat{y} \in \{A, B, C, D\})$—the standard approach in calibration research [Kadavath et al., 2022]. The primary outcome is the *signed calibration gap*, $\text{gap}_i = c_i - y_i$ with $y_i \in \{0, 1\}$ indicating correctness; it is positive under overconfidence, negative under underconfidence, and supports regression-based inference that binning-based ECE cannot. Note that since $y_i \sim \text{Bernoulli}(p_j)$, residual variance is highest near 50% accuracy, so the homoskedasticity assumption of the mixed model below is imperfect.

Question-level covariates. We extract four features from each question, all standardized before fitting: **word count** of the question stem; **max option length** (word count of the longest answer choice); **negation indicator** (*not*, *except*, *least*, *never*, *no*); and **entropy** of the softmax distribution over A/B/C/D in nats, $H_i = -\sum_k p_{ik} \log p_{ik}$, ranging from 0 to $\log 4 \approx 1.386$.

Elicitation judgment calls. We extract confidence from token-level logprobs over {A, B, C, D} rather than asking the model to verbalize a probability, for two reasons. First, logprob-based elicitation is model-agnostic and avoids conflating calibration with instruction-following ability—a model that is poorly calibrated may still correctly say "I am 80% confident" if trained to do so. Second, we softmax-renormalize over the four answer tokens rather than the full vocabulary, which conditions on the assumption that one of the four options is correct and yields a proper probability distribution over choices. We pre-fill the assistant turn with "The answer is" to force scoring at the next-token position, following Kadavath et al. [2022]; an alternative would be scoring across the full completion, but this is less tractable and model-dependent.

4.2 Nonparametric Calibration Curves

For each subject we estimate a reliability curve using isotonic regression [Zadrozny and Elkan, 2002] rather than equal-width binning, with an unconstrained Gaussian kernel smoother ($\sigma = 1.5$ bins) overlaid as a monotonicity-assumption check. ECE per subject is the weighted mean absolute deviation between the reliability curve and the diagonal, using 20 equal-width confidence bins.

[1]The fp16 1.5B run was added after we observed that the 4-bit 1.5B run had unusual positional bias (predictions skewed toward B at 36%) and accuracy 0.054 below the published full-precision figure. The fp16 run brought both numbers in line with the published figures and is used as the canonical 1.5B in the main analysis. The 4-bit version is retained for reference.

4.3 Multilevel Regression and Variance Decomposition

We model the question-level calibration gap across all 14,042 questions using a three-level mixed model with questions nested in subjects nested in domains:

$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^{\top}\boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \tag{1}$$

where $\mathbf{x}_{ijk}$ are standardized question-level covariates, $u_k \sim \mathcal{N}(0, \sigma^2_{\text{domain}})$ is a domain random intercept, $v_{jk} \sim \mathcal{N}(0, \sigma^2_{\text{subject}})$ is a subject-within-domain random intercept, and $\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma^2_{\varepsilon})$ is residual noise. The model is estimated by restricted maximum likelihood (REML) using `statsmodels.MixedLM` [Raudenbush and Bryk, 2002].

The fixed effects $\boldsymbol{\beta}$ characterize which question features are associated with miscalibration across the full dataset, interpreted as descriptive associations rather than causal effects. The variance components $(\sigma^2_{\text{domain}}, \sigma^2_{\text{subject}}, \sigma^2_{\varepsilon})$ answer RQ1 via an intraclass correlation (ICC) decomposition: the fraction of total calibration heterogeneity attributable to each level of the hierarchy.

Modeling judgment calls. Several design choices warrant explicit justification. **Fixed vs. random effects at each level.** Question-level features are modeled as fixed effects because we seek population-level associations between item characteristics and miscalibration—a single coefficient per feature applying across all questions and subjects. Treating these as random slopes by subject would estimate 57 separate coefficients per feature, which is neither the scientific question nor warranted given average subject sizes of ∼246 questions. Subjects are modeled as random effects rather than fixed because we are interested in the variance they explain, not their individual intercepts, and because random effects enable the ICC decomposition central to RQ1. The exchangeability assumption—that subjects are draws from a population of academic knowledge domains—is reasonable given MMLU's design. Domains are also modeled as random effects: with only four groups, fixed effects would estimate specific domain intercepts at the cost of a degree of freedom each, while random effects treat domain as a source of variance to be estimated and tested. **Three levels vs. two.** We include a domain random intercept because MMLU's four-domain hierarchy is the natural grouping structure; in practice domain variance collapses to zero (Table 1), confirming a two-level model suffices, but the three-level specification is the appropriate starting point. **Signed gap as outcome.** We model the signed calibration gap rather than ECE directly because it supports regression-based inference; ECE is derived separately per subject from isotonic curves.

4.4 Multiple Testing and Cross-Model Replication

For each of the 57 subjects we test H_0: mean calibration gap = 0 via a two-sided one-sample t-test on the question-level gaps, controlling the false discovery rate at $\alpha = 0.05$ via Benjamini–Hochberg [Benjamini and Hochberg, 1995]. The full pipeline is run on each of the seven models, and subject-level ECE rankings are compared across all 21 model pairs with Spearman rank correlation and bootstrap CIs (2000 replicates, seed 42).

5 Results

We organize results around the three research questions: Sections 5.1–5.3 establish where within MMLU calibration concentrates (RQ1), Section 5.4 tests whether these patterns are stable across models (RQ2), and Section 5.5 examines what question-level and scale-related mechanisms produce the heterogeneity (RQ3). Section 5.6 reports sensitivity checks on all findings.

5.1 Variance Decomposition (RQ1)

Table 1 reports the ICC decomposition from the three-level mixed model for the seven main models. Across all seven, the bulk of calibration-gap variance lies at the question level (92.0–98.5%), and domain-level variance collapses to zero. Between-subject ICCs span nearly a factor of five: GPT-4o-mini concentrates 7.9% of variance at the subject level, Qwen-14B 5.9%, Qwen-3B 5.1%, Qwen-7B 4.9%, and Llama and the smaller Qwen models cluster around 1.7–2.0%.

Table 1: Variance decomposition of the calibration gap. Domain-level variance is estimated at zero for every model; $\mathrm{ICC}_{\mathrm{subject}}$ is the share of total variance attributable to subject-level heterogeneity.

Model	$\sigma^2_{\mathrm{domain}}$	$\sigma^2_{\mathrm{subject}}$	σ^2_{ε}	$\mathrm{ICC}_{\mathrm{subject}}$
GPT-4o-mini	0.000	0.013	0.150	0.079
Llama-3-8B-Instruct	0.000	0.004	0.190	0.020
Qwen-2.5-0.5B-Instruct	0.000	0.004	0.224	0.018
Qwen-2.5-1.5B-Instruct	0.000	0.003	0.197	0.017
Qwen-2.5-3B-Instruct	0.000	0.011	0.203	0.051
Qwen-2.5-7B-Instruct	0.000	0.009	0.177	0.049
Qwen-2.5-14B-Instruct	0.000	0.010	0.156	0.059

The within-Qwen scaling sweep across four matched-precision checkpoints (1.5B fp16, 3B 4-bit, 7B 4-bit, 14B 4-bit) shows ICC making a substantial jump from 1.5B to 3B (0.017 $\rightarrow$ 0.051) and remaining elevated at the larger scales (0.049 at 7B, 0.059 at 14B). The discontinuity around 1.5B$\rightarrow$3B suggests that subject-level concentration of miscalibration emerges around 2–3B parameters, then increases gradually; combined with the parallel monotonic increase in the entropy coefficient (Section 5.5), this is the cleanest within-family scaling signal in our data.

The domain-level variance is estimated at zero for every model. With only four domain groups, the likelihood is flat with respect to the domain variance component, and the estimate collapses to the boundary. We therefore report total between-subject variance without a clean domain/subject partition, and treat domain as a descriptive grouping rather than a modeled random effect.

5.2 Calibration Curves by Subject

Figure 1 contrasts reliability curves for the five most and five least miscalibrated subjects (by ECE) for GPT-4o-mini. Curves are masked outside the observed 1st–99th percentile of confidence per subject to avoid displaying isotonic extrapolation: GPT-4o-mini's confidence is so concentrated above 0.7 that any plotted curve below this region would be pure extrapolation. Within the observed support, the worst-calibrated subjects show the characteristic shape of an overconfident model—empirical accuracy lying below the diagonal, sometimes sharply—while the best-calibrated subjects track the diagonal closely. Both shapes are present in the same model on the same evaluation, illustrating the heterogeneity that aggregate ECE conceals.

5.3 FDR-Corrected Subject Rankings

Figure 2 shows all 57 subjects sorted by mean calibration gap for each of the seven main models, with FDR-rejected subjects highlighted. At $\alpha = 0.05$: GPT-4o-mini, Llama, Qwen-3B, Qwen-7B, and Qwen-14B reject all 57 subjects; Qwen-1.5B (fp16) rejects 56/57 (the lone exception is `jurisprudence`, $\bar{\mathrm{gap}} = 0.051$, $p_{\mathrm{adj}} = 0.18$); Qwen-0.5B rejects 55/57 (the exceptions are `high_school_statistics` and `formal_logic`).

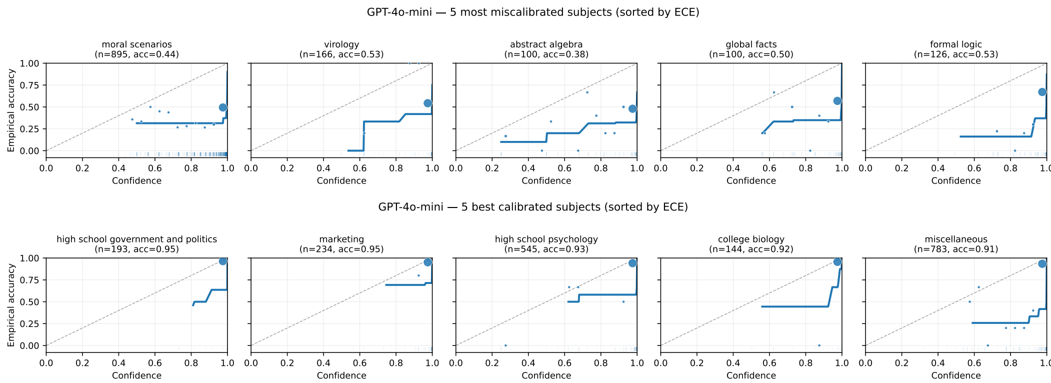

Figure 1: GPT-4o-mini reliability curves. **Top**: five most miscalibrated subjects by ECE. **Bottom**: five best-calibrated subjects. Same conventions in both: solid line is isotonic regression, plotted only on observed support (1st–99th percentile per subject); markers are 20-bin empirical accuracy sized by bin count; dashed line is the calibration diagonal.

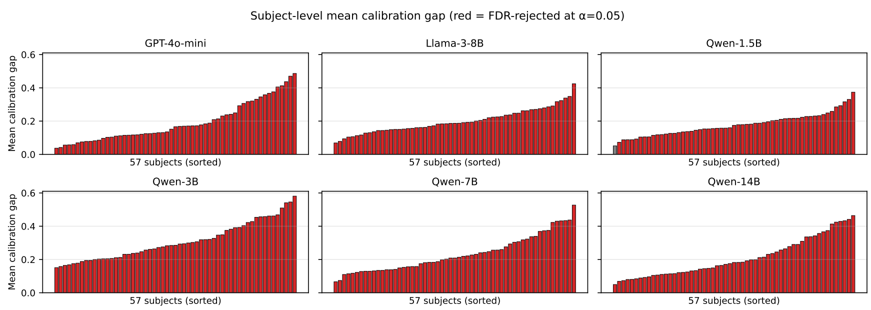

Figure 2: Mean calibration gap per subject, sorted ascending, for each of the seven main models. Red bars indicate subjects where H_0: mean gap $= 0$ is rejected at FDR $\alpha = 0.05$; gray bars are not rejected.

The substantive finding is the *magnitude* of heterogeneity, not the binary reject outcome: best- to worst-calibrated gap ranges span 3.8–13.0× across the six larger models, and the *same* subjects sit at each extreme.

Subjects that are robustly worst-calibrated. Across the six larger models (excluding Qwen-0.5B due to its positional-bias confound), two subjects appear in the top-10 most miscalibrated for every model above 1B parameters: `virology` and `professional_law`. A consistent cluster of college-level technical subjects—`abstract_algebra`, `college_mathematics`, and `machine_learning`—each appears in the top-5 most miscalibrated for four of the six larger models. Together these form two recurring patterns: (i) college-level quantitative and technical content and (ii) niche professional or factual knowledge (`virology`, `professional_law`).

Subjects that are robustly best-calibrated. Symmetrically, `marketing` and `high_school_psychology` appear in the bottom-10 best-calibrated for every model above 1B parameters, and `high_school_government_and_po` is consistently among the best-calibrated across models. Two patterns are visible. First, MMLU's high-school-level subjects tend to show smaller calibration gaps than their college-level counterparts—the level of specialization required appears to track miscalibration. Second, the bottom-10 intersection is dominated by Social Sciences and the "Other" (applied/factual) MMLU category, with no STEM subject appearing.

5.4 Cross-Model Agreement (RQ2)

Figure 3 shows the 7×7 Spearman rank correlation matrix for subject-level ECE across all seven main models. Among the six larger models, all 15 pairwise correlations fall in $[0.80, 0.94]$. The within-family pair Qwen-7B vs Qwen-14B is the highest at 0.94; cross-family GPT-4o-mini vs Qwen-7B is 0.92 and GPT-4o-mini vs Qwen-14B is 0.91.

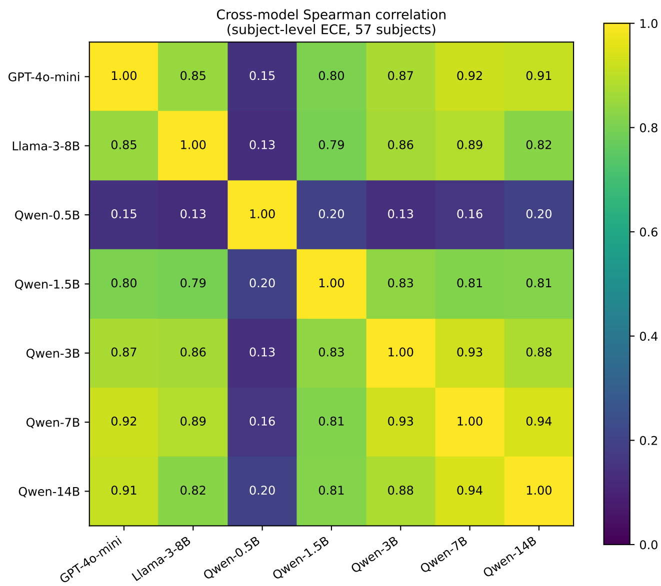

Figure 3: Spearman rank correlation of subject-level ECE across the seven main models (57 common subjects). Among the six larger models, all 15 pairwise correlations fall in $[0.80, 0.94]$. Qwen-0.5B's correlations with the others are $[0.13, 0.20]$, reflecting a qualitatively different miscalibration pattern driven by extreme positional bias.

Qwen-0.5B's correlations $[0.13, 0.20]$ (bootstrap CIs include zero) reflect its 10.6× positional bias toward D rather than genuine knowledge gaps; we read this as evidence that positional bias dominates the calibration signal below approximately 1B parameters in our elicitation protocol.

5.5 Predictors of Miscalibration (RQ3)

Table 2 reports the fixed effects from the multilevel model across the seven main models. Intercepts confirm large mean overconfidence in every model (0.179–0.311). Within the Qwen family, the intercept is *non-monotonic* in scale: it rises sharply from 1.5B (0.179) to 3B (0.311), then declines back at 7B (0.233) and 14B (0.210). Mean accuracy is monotonic ($0.571 \rightarrow 0.601 \rightarrow 0.685 \rightarrow 0.757$), so the inverted-U intercept reflects the confidence outpacing accuracy at intermediate scale, then accuracy catching up at larger scales.

Figure 4 shows the corresponding forest plot. Word count is a positive significant predictor for GPT-4o-mini and the larger Qwens (7B, 14B), inert for Llama and the small Qwens (0.5B, 1.5B, 3B). The most informative pattern is the entropy coefficient.

Table 2: Fixed effects from the three-level mixed model for all seven models. Continuous covariates standardized. Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.

Predictor	GPT	Llama	Q-0.5B	Q-1.5B	Q-3B	Q-7B	Q-14B
Intercept	0.196***	0.197***	0.209***	0.179***	0.311***	0.233***	0.210***
Word count	0.031***	-0.005	-0.013*	0.003	0.010	0.019**	0.023***
Max option length	0.001	-0.001	-0.028***	0.005	-0.008†	-0.007	-0.007
Negation	-0.001	0.007	0.016	-0.010	-0.014	-0.001	0.005
Entropy	0.016***	0.008†	-0.063***	-0.004	0.006	0.029***	0.040***

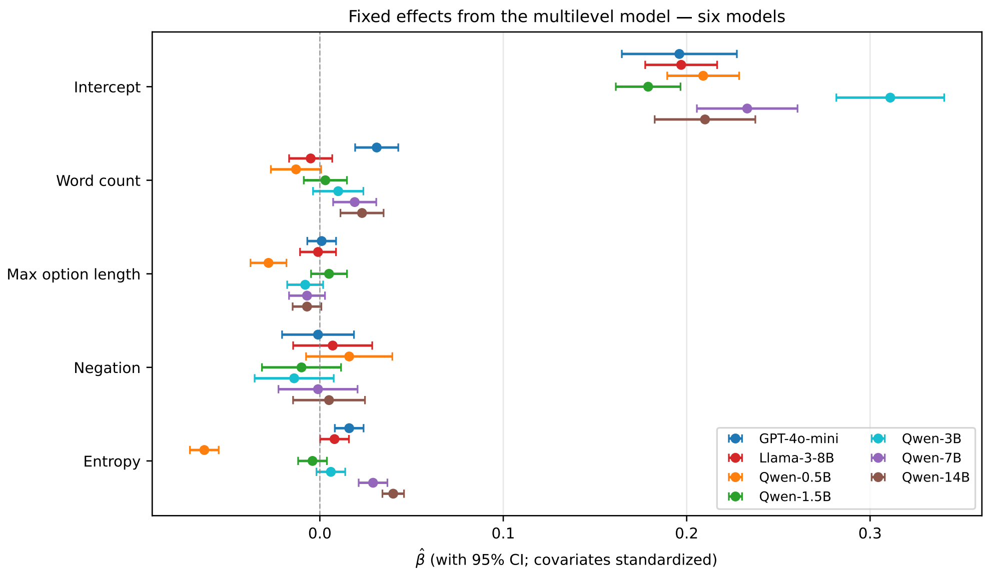

Figure 4: Fixed effect estimates with 95% confidence intervals for all seven models. Estimates are on the standardized calibration gap scale. The entropy coefficient shows a striking sign progression across the Qwen scaling sweep.

Entropy: a sign progression with scale. A naïve prediction is that high option-distribution entropy reduces the calibration gap, since top-1 confidence is mechanically lower when probability mass spreads across options. Yet the entropy coefficient varies systematically across models and scales: strongly negative for Qwen-0.5B (-0.063, $p < 0.001$), near-zero for Qwen-1.5B and Qwen-3B (-0.004 and $+0.006$, both n.s.), and positive and growing across larger models (Llama $+0.008$, GPT-4o-mini $+0.016$, Qwen-7B $+0.029$, Qwen-14B $+0.040$, all $p < 0.10$ or smaller). Within the Qwen family at matched precision (1.5B fp16, 3B, 7B, 14B), the coefficient grows monotonically across all four scale points: $-0.004 \to +0.006 \to +0.029 \to +0.040$. Figure 5 traces the mechanism. Within each model we partition questions into entropy quartiles and report mean confidence (solid) and mean accuracy (dashed); the shaded area is the calibration gap.

Note that confidence does fall with entropy (the max probability is lower on high-entropy items), but accuracy falls faster, so the calibration gap widens on the questions the model finds hardest. Within the Qwen family the pattern is one of confidence growing faster than accuracy as items become harder, rather than accuracy collapsing on hard items: at 1.5B confidence and accuracy move roughly together across quartiles (gap $+0.09$ at Q1, $+0.13$ at Q4), but at 7B and 14B confidence remains near saturation across the first three quartiles while accuracy still drops with entropy (Q1

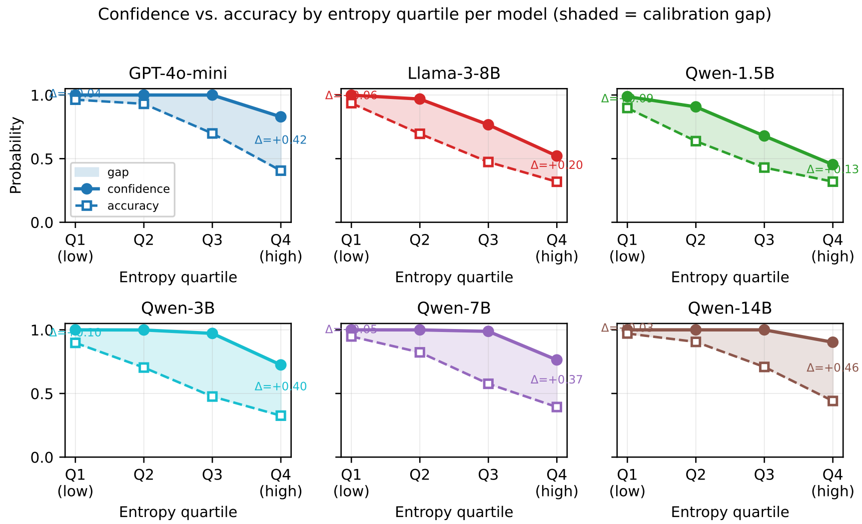

Figure 5: Confidence (solid) and accuracy (dashed) by entropy quartile per model, with the shaded gap. The pattern progresses systematically across the Qwen family: at small scale (0.5B) the gap shrinks with rising entropy; at intermediate scale (1.5B) the curves fall in parallel; at larger scales (7B, 14B) the gap widens with entropy as confidence stays near saturation across the first three quartiles while accuracy still falls with entropy.

vs. Q4 gaps: +0.05 vs. +0.37 at 7B; +0.03 vs. +0.46 at 14B). This is consistent with prior reports of RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026]. We treat the within-family comparison as observational—four checkpoints, one family—and read Qwen-0.5B's strongly negative entropy coefficient as an artifact of its positional bias rather than a fourth scale point.

5.6 Sensitivity Analysis

Sensitivity-analysis conclusions extend uniformly: ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. 20-bin baseline), the Negative Log Calibration Score (a strictly proper scoring rule) ranks subjects nearly identically (Spearman ≥ 0.79), and two-level vs. three-level ICC estimates are numerically identical (consistent with $\sigma^2_{\text{domain}} = 0$). The FDR rejection is also robust to threshold: at $\alpha = 0.01$, the four largest models reject 55–57/57 and the smaller Qwens 46–55/57 (the 4-bit 1.5B run is the most sensitive to threshold choice).

6 Discussion

Where calibration succeeds versus fails. The two strongest findings are descriptive but coherent. Calibration is reliably good on high-school-level Social Sciences and applied/factual subjects (`marketing`, `high_school_psychology`, `high_school_government_and_politics`), and reliably bad on college-level quantitative and technical content (`abstract_algebra`, `college_mathematics`, `machine_learning`) and on niche professional and factual knowledge (`virology`, `professional_law`). The same patterns appear in every model larger than 1B parameters that we examined: all 15 pairwise Spearman correlations among the six larger models fall in $[0.80, 0.94]$, with cross-family GPT-4o-mini vs Qwen-7B/14B at 0.92/0.91 comparable to the strongest within-family pairs. Subject-level miscalibration is therefore primarily a property of the task, providing empirical motivation for scoping corrections at the subject level rather than globally—though each correction would still be fit per-model on held-out data.

Within-family scale and implications. Within the Qwen family at matched precision (1.5B fp16, 3B, 7B, 14B) the entropy coefficient grows monotonically with scale, the between-subject ICC jumps from 1.5B to 3B then plateaus, and the intercept peaks at 3B before declining—capable models maintaining near-saturated confidence as items get harder, consistent with prior reports of RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026]. We treat this as observational over four checkpoints. The cross-model stability is the precondition that makes subject-aware recalibration worth scoping; the recurring offenders (`professional_law`, `virology`, college-level technical content) are a natural starting point for pre-deployment review on new models without re-running the full 57-subject audit.

Limitations. With only four domain groups, domain-level variance is estimated at zero and cannot be cleanly partitioned from subject-within-domain. The mixed model assumes homoskedastic residuals (violated near 50% accuracy); confidence is a single-token logprob, which may not represent internal uncertainty after RLHF [Kadavath et al., 2022]; and results are specific to MMLU's four-choice format. The within-family scaling comparison is observational, and Qwen-0.5B contributes no usable signal due to extreme positional bias. 4-bit quantization affects five of seven models; a 1.5B 4-bit-vs-fp16 comparison showed accuracy $0.536 \to 0.571$ and FDR rejection $46/57 \to 55/57$ at $\alpha = 0.01$, suggesting quantization depresses confidence at the smallest scales, but we did not repeat the comparison at larger scales.

Future directions. Additional Qwen checkpoints (e.g., 32B), a fourth family (Mistral or Gemma) at matched scale, and replication on MMLU-Pro would all strengthen the generalization claim.

7 Conclusion

Subject-level calibration heterogeneity in LLMs is large and stable across model families above approximately 1B parameters: applied/factual high-school subjects are reliably well-calibrated, while college-level quantitative and technical content and niche professional and factual knowledge are reliably bad. Subject-level ECE rankings correlate strongly across all 15 pairs of larger models (Spearman $\rho \in [0.80, 0.94]$), pointing to task structure rather than any particular training pipeline. Within the Qwen family the entropy coefficient grows monotonically with scale, in line with prior reports of RLHF-induced confidence saturation. The audit's stable targets provide empirical motivation for subject-aware recalibration scoping and a transferable diagnostic for flagging miscalibration on new models.

Statement of Work

Data analysis. Jennifer conducted the original two-model analysis (GPT-4o-mini, Llama-3-8B-Instruct), including the multilevel model, calibration curves, FDR correction, and sensitivity checks. John collected and analyzed the six additional Qwen-2.5 model runs (0.5B, 1.5B 4-bit, 1.5B fp16, 3B, 7B, 14B), extended the multilevel and cross-model framework to seven models, and produced the within-family scale, entropy-mechanism, and quantization-effect analyses. **Coding.** Jennifer built the original pipeline; John extended it to support arbitrary models and produced the parameterized inference notebook used for the open-weight models. **Writing.** Jennifer and John jointly drafted the report with the seven-model expansion. **Presentation.** Jennifer and John jointly prepared the slide deck.

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